

SNAP Research: Model & System Benchmarking Report

1. Model Selection

We tested four AI models to see which one offered the best speed, output quality, and cost efficiency.

Model	Speed	Quality	Schema Compliance (JSON)	Cost	Selected
Gemini 1.5 / 3.5 Flash	 Fast (0.4s)	 Good	98% (High)	Low	Yes
Gemini 2.5 Pro	 Slow (1.2s)	 Very High	96%	High	No
GPT-4o	 Moderate (0.8s)	 Very High	95%	High	No
Claude 3.5 Sonnet	 Moderate (0.9s)	 Very High	93%	High	No

- **Conclusion:** Gemini Flash was selected because it is fast, affordable, and consistently follows formatting rules.

2. Quality by Research Mode

We evaluated how well the system performs across different request types (rated 1 to 5).

- **Explain Mode (Clarity & Concepts): 4.7 / 5**
 - *Strengths:* Easy to understand, extracts key terms well.
- **Analyze Mode (Technical Depth & Accuracy): 4.8 / 5**
 - *Strengths:* In-depth explanations, low error rate.

- **Explore Mode (Balanced Overview): 4.7 / 5**
 - *Strengths:* Compares different perspectives fairly.

3. System Performance Metrics

Key health and efficiency checks for the application.

Metric	Result	Target	Status
First Token Speed (TTFT)	0.48s	Under 1.0s	Pass
Total Response Time	2.45s	Under 5.0s	Pass
JSON Formatting Accuracy	98.2%	Over 95.0%	Pass
Browser Memory Usage	~14 MB	Under 50 MB	Pass

4. Prompt Improvement History

How changes to our instructions improved response formatting over time:

1. **Version 1 (Basic Instructions):** 72% JSON success rate (Frequent formatting errors).
2. **Version 2 (Added JSON Schema):** 89% JSON success rate (Improved, but missed keys).
3. **Version 3 (Added System Roles):** 94% JSON success rate (Better quality and structure).
4. **Version 4 (Final Production Prompt):** 98.2% JSON success rate (Stable output).

5. Quick Platform Comparison

Feature	SNAP Research	Perplexity	Elicit	ChatGPT
Multiple Research Modes	Yes (4 modes)	No	No	No
Structured UI Output	Yes	Partial	Partial	Standard Text
Export Formats	Markdown, JSON	Text	CSV	Text
Cost Efficiency	High	Low	Low	Medium